

Week-1:

C-Programming: Basic Input, Operators And conditional statements.

Lab Report

1. Compulsory programs :-

1. To check whether the number is even or odd:
 - Start the program.
 - Read an integer from the user.
 - Find the remainder when the number is divided by 2.
 - If the remainder is 0, display "It is an even number" or display "It is an odd number".
 - End of the program.

2. To check whether the number is positive, Negative or zero.
 - Start the program.
 - Read an integer from the user.
 - Compare the number with 0.
 - If it is greater than '0' display "It is a positive number", if it is less than '0' display "It is a negative number" otherwise print "It is zero".
 - End of the program.

To check whether a given character is an uppercase letter, lowercase letter, digit or special character.

- Start the program.
- Read the character from the user.
- Using else if ladder, check the character using logical AND operator.
- If the character is 'A' to 'Z' display "It is upper case", if the character is 'a' to 'z' display "It is lower case" & if the character is '0' to '9'

display "It is a digit" or display "It is a special character".

→ End of the program.

4. To check whether the given year is leap year or not.

→ Start the program.

→ Read the input from the user.

→ If it is divided by four, then the remainder either '0' or something.

→ If the remainder is '0' display "It is a leap year" otherwise print "It is not a leap year".

→ End of the program.

5. To print memory allocation of all types of data in C.

→ Start the program.

→ Using int, char, float, unsigned int and signed int, etc. as datatypes.

→ With the help of size of () operator. Then it displays the memory allocation of all datatypes.

→ End of the program.

II. Medium programs :-

1. To print arithmetic operations using switch statement.

→ Start the program.

→ Take a, b and choice as variables.

→ Read the input from the user as two operands.

→ Now, print a menu of arithmetic operations like Addition, Subtraction, Multiplication, etc. -

→ Enter your choice and the operation will be done using switch case statement.

- Then the output was given.
- End of the program.

2. Program to swap two given numbers without using an extra variable.

- Start the program.
- Take a, b as variables to store data.
- Read input from the user as two numbers.
- Now, using arithmetic operations, swap the two values of a & b . Then the values will be swapped.
- Then it prints the values of a & b after swapping.
- End of the program.

3. Program to swap two given numbers using a temporary variable.

- Start the program.
- Take a, b, c as variables to store data.
- Read the input from the user as two numbers.
- Now using third variable as c , swap the values of two variables.
- Then it prints the values of a & b after swapping.
- End of the program.

4. A program to check the given number is perfect square.

- Start the program.
- Take num, i and initialize the value $p=0$ as variables.
- Read the input from the user.
- Using for loop initialize the value $i=1$, give the condition and then increment should be given in the input.
- First it reads the number and checks the condition.

then in each iteration, calculate $i \times i$ and compare it with num.

→ If $i \times i$ is equal to num, then, increment p by 1 and terminate the loop using break statement.

→ After that loop, check whether $p=1$. If it is $p=1$, display "It is a perfect square" otherwise display "It is not a perfect square".

→ End of the program.

5. Program to read a student's marks and print grades.

→ Start the program.

→ Take marks as a variable to store data.

→ Read the input from the user.

→ Using Else if statement and using logical AND operators, comparative operators.

→ Give the condition as per marks and it gives the corresponding grades.

→ Otherwise print "Fail".

→ End of the program.

11. Advanced programs :-

1. Program to print arithmetic operations using the default case and division by zero using a conditional check.

→ Start the program.

→ Take a, b, choice as variables to store data.

→ choice is to enter your choice and print the menu of arithmetic operations.

→ Using switch case print the addition, subtraction, multiplication, etc. operations.

→ Now, using division by zero, print if it is $b! = 0$ and ^{otherwise} print "Division by zero is not possible".

→ In default, enter "Invalid choice" and gives the output.

→ End of the program.

2. Program to find the largest of three given numbers only the conditional (ternary) operator.

→ start the program.

→ Take a, b, c, largest as variables to store the data.

→ Using ternary operator, compare the three given numbers which were taken from the user.

→ Then it prints the output of largest of three given numbers.

→ End of the program.

3. Program to convert a numeric grade (0.0 to 4.0) into a letter grade +/- using switch statement.

→ start the program.

→ Read Grade point and which is taken from the user.

→ Now using switch case statement print 'A' if $dec > 0.0$ & $dec < 0.5$ otherwise print "A-", like that print the grades.

→ It checks all the conditions which were given in switch statements and then if the condition was wrong then it goes to another condition.

→ Then the default statement will display, if the user entered an invalid input.

→ End of the program.